HS Data Science 26

Session 05: How to Give a Talk

Prof. Dr. Michael Granitzer

2 June 2026

Session Overview

The Fundamentals

- a talk is not a paper — different purpose, different rules
- who is your actual audience?
- what should they leave with?

What to Put In

- motivation: the 20% that earns the other 80%
- the key idea: narrow and deep, not wide and shallow
- examples as the main weapon
- what to leave out (almost everything)

How to Structure

- the roller-coaster model
- title → motivate → key idea → details → conclusion
- finishing on time is not optional

How to Present

- enthusiasm is contagious (and so is nervousness)
- being seen and being heard
- the jelly effect and how to fight it
- questions as an opportunity, not a threat

> A talk is an **advertisement** for your paper or idea. The goal is not to transmit all information — it is to make the audience want to know more.

Talk vs. Paper

> **Guiding Question:** What does a research talk achieve that a paper cannot?

The Fundamental Difference

Source: Peyton Jones 2016; Peyton Jones 2016

A paper is the beef.

A paper contains all the evidence: proofs, tables, confidence intervals, related work, appendices. Readers set their own pace. They can re-read difficult passages.

A talk is the advertisement.

A talk exists to:

1. give an intuitive feel for what was done
2. make the audience eager to read the paper
3. help people remember that this work exists

> You will not convey all the information in your paper. This is not a failure. It is the design.

What follows from this

Paper	Talk
All results	Key result only
Full related work	One motivating comparison
Proofs / derivations	Intuition only
Error bars on every table	The headline number
Limitations section	One honest caveat
8–15 pages	20–30 slides

The most common mistake

Trying to compress the entire paper into the time slot.

The talk then becomes a very fast, very hard-to-follow paper — and the audience switches off.

Who Is Your Actual Audience?

The imagined audience

When preparing, speakers tend to imagine their ideal audience: experts in the exact sub-field, who have read the relevant papers, and who are paying close attention.

The actual audience

Most people in the room:

- work in adjacent but different areas
- did not read the paper being presented
- are not fully concentrating (jet lag, hunger, email on phone)
- will remember at most one or two things

The implication

Design for the actual audience, not the imagined one. If you design for the imagined audience, you lose the actual one.

Calibrating to the room

Before the talk, ask:

- How expert is this audience in my specific topic?
- How much background can I assume?
- What do they care about that my work addresses?
- What jargon must I define?

Different settings require different talks

Setting	Audience calibration
Lab group seminar	expert, detailed, open discussion
Conference (e.g. ACL)	expert in field, but not your sub-topic
Departmental colloquium	mixed expertise, wider framing needed

Setting	Audience calibration
Industry talk	no paper norms; focus on impact and application
Seminar course	peer students — be pedagogical, not impressive

What to Put In

Guiding Question: What content earns audience attention — and what content loses it?

The 20% Rule: Motivation

The Key Idea — Narrow and Deep

Choose one idea

Every talk has (or should have) one central idea. One. Not three.

If you have three contributions, one of them is the talk. The others are supporting evidence or future work.

Narrow and deep beats wide and shallow

Wide and shallow: "We improved retrieval, generation, ranking, and evaluation for RAG systems."

Narrow and deep: "We show that the bottleneck in RAG is not retrieval quality, but the reader's attention window — and we fix it with a single architectural change."

The second version has a claim. The first has a list.

≡

Developing the key idea

Ask yourself: if the audience remembers only one sentence from my talk, what should it be?

Write that sentence down. Put it on slide 2. Return to it at the end.

The key idea slide should show:

1. The core claim in plain language
2. A figure or example that makes it intuitive
3. The headline result that supports it

What to omit

- all secondary contributions
- ablation tables (unless they directly support the core claim)
- implementation details (reserve for questions)
- related work beyond what establishes the gap

> If you cannot state your key idea in one sentence, you have not understood it well enough to present it.

Examples Are Your Main Weapon

Why examples work

- they are concrete: the audience processes them immediately
- they create memory: people remember stories, not abstractions
- they demonstrate understanding: showing a good example proves you know the domain
- they are faster than formalism: one worked example can replace a page of derivation

Examples in AI/ML talks

- show a failure case of the baseline and a success case of your method, side by side
- use a toy dataset to build intuition before showing full results
- trace a single data point through your architecture
- show an attention map, a generated sentence, a confusion matrix — something visual and concrete

How to choose your example

Choose an example that:

- is simple enough to understand in 30 seconds
- is representative (not a cherry-pick)
- shows the key failure mode of prior work
- is clearly fixed by your approach

Examples you should never use

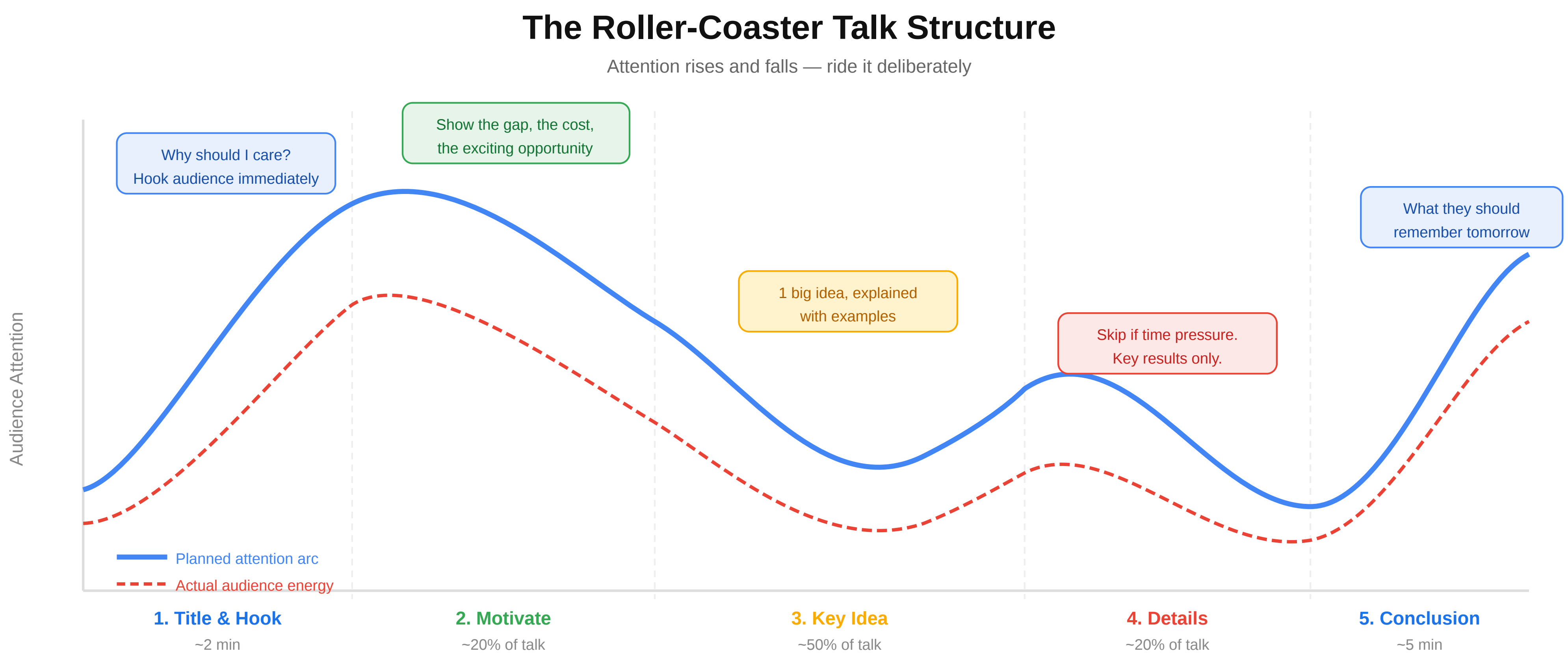
- examples that require domain expertise to appreciate
- examples that are technically impressive but visually dull
- examples that need two slides of setup to understand
- examples that make your method look good but are obviously unrealistic

For every abstract claim in your talk, have a concrete example ready. If you cannot find one, reconsider whether the claim belongs in the talk.

How to Structure

> **Guiding Question:** What structure keeps the audience engaged from the first slide to the last?

The Roller-Coaster Model



Section by Section

1. Title & Hook (~ 2 min)

- state the title and your name
- immediately follow with a hook: a question, a striking result, a demo
- do not give an outline yet — earn attention first

2. Motivate (~ 20% of talk)

- the problem, its cost, why it is hard
- one comparison to existing approaches
- end with: "Our approach addresses this by…"

3. Key Idea (~ 50% of talk)

- the core contribution, explained intuitively
- at least two worked examples
- the headline result
- one clear figure that summarises the method

4. Details (~ 20% of talk, skippable)

- selected technical depth for the experts in the room
- one table of results (highlight the key number only)
- an ablation if it directly supports the key claim
- **skip or compress if running late — never rush the conclusion**

5. Conclusion (~ 5 min)

- restate the key idea in one sentence
- the one result the audience must remember
- implications and open questions
- "Thank you" — and stop

> Finishing on time is a matter of professional respect. **Always rehearse.** A talk that runs

≡

over is a talk that disrespects the audience and the schedule.

How to Present

> **Guiding Question:** What separates a forgettable delivery from one that leaves an impression?

Enthusiasm Is Contagious

The most important rule

If you are not enthusiastic about your work, no one else will be.

Enthusiasm is not about being loud or theatrical. It is about:

- speaking at a pace that sounds engaged (not rushed, not flat)
- using your voice to emphasise what matters
- showing genuine curiosity about the questions your work raises
- smiling when something is surprising or elegant

The jelly effect

Nervousness causes: monotone voice, reading from slides, avoiding eye contact, speaking too fast, fidgeting.

Nervousness is normal. Preparation is the cure.

Practical cures for nervousness

1. **Rehearse out loud** — three times minimum. Reading silently does not work.
2. **Memorise the first 60 seconds** — if the opening is fluent, confidence builds naturally.
3. **Slow down by 20%** — nerves make you speak faster. Intentional slowing reads as confidence.
4. **Look for nodders** — find one person in the audience who is engaged and nodding. Make eye contact with them. This anchors you.
5. **Welcome questions as interest** — a question means the audience is engaged. An empty Q&A is the worst outcome.

> You will never feel perfectly ready. Give the talk anyway. The only cure for nervousness is

repetition.

Being Seen and Being Heard

Voice

- speak to the **back row**, not the screen
- project from the diaphragm, not the throat
- vary pace and pitch — monotone loses attention within minutes
- pause before key points — silence commands attention
- never read slides word for word

Body

- stand still or move with purpose — don't pace randomly
- face the audience, not the screen
- use the laser pointer sparingly (it draws attention away from your voice)
- gestures should reinforce, not distract

Eye contact

- scan the whole room, spending 3–5 seconds on each person
- when making a key point, hold eye contact with one person for the full sentence
- find nodders early; return to them when you need anchoring

The room

- arrive early, check the projector and audio
- know where the clicker advance button is before you start
- if a slide fails, narrate around it — never apologise for technical issues
- manage the room temperature with the organisers in advance if possible

> The audience watches the speaker, not the slides. Design your presence as carefully as your slides.

Handling Questions

Questions are an opportunity

A question means the audience was engaged. A tough question means an expert was interested enough to push back. Both are success signals.

Common question types

Type	How to respond
Clarification	"Great question — let me re-explain X"
Extension	"That would be interesting future work"
Challenge	"That's a fair concern; here's what we found…"
Off-topic	"Happy to discuss offline — in the interest of time…"
Unknown	"I don't know — let me think about it and follow up"

Rules for Q&A

1. **Listen to the full question** before starting to answer
2. **Repeat or rephrase** for the rest of the audience ("So the question is…")
3. **It is fine to say "I don't know"** — much better than guessing
4. **Keep answers short** — one question should get one answer, not a mini-lecture
5. **Watch the chair** — when they say time is up, stop, even mid-sentence

After the talk

- follow up with anyone who asked an interesting question
- note difficult questions for the paper revision or future work section
- questions reveal what the audience misunderstood — redesign those slides for next time

Slide Design

Guiding Question: What makes a slide help the audience — rather than distract them?

Slide Principles

Less is more

- one idea per slide
- maximum 5–6 bullet points, each no longer than one line
- large font (minimum 24pt for body text)
- high-contrast colours (dark text on light background, or vice versa)
- no more than 1 slide per minute for a fast-paced technical talk

Figures beat tables

- a good figure conveys a trend in 5 seconds
- a table requires 30 seconds minimum to read
- if you must show a table, highlight the one number that matters

Slide titles are claims, not topics

Bad: "Results"

Things to avoid

- full sentences copied from the paper
- reading the slide aloud word for word
- animations that add no information (spinning, bouncing)
- clip art or stock photos unrelated to the content
- tiny fonts on dense figures from the paper — redraw at talk scale
- "I know you can't read this, but…" — then don't show it

Useful slide types

Type	When to use
Single claim + figure	core idea slides
Before / after	showing improvement

Good: "Our method improves MRR@10 by 8% on all datasets"

Type	When to use
Timeline	contextualising the work
Single equation	one key formula only
Demo screenshot	system / application work

> Slides are a visual aid, not a teleprompter. Your voice carries the content; slides provide the anchor.

Summary

Core rules

1. A talk is an advertisement — not a compressed paper
2. Design for the actual audience, not the imagined one
3. Spend 20% on motivation — earn attention before explaining
4. One key idea, explained narrow and deep
5. Examples are your main weapon — concrete always beats abstract
6. Finish on time — rehearse three times out loud

Structure

Title & Hook → Motivate → Key Idea → Details (optional) → Conclusion

Presentation

- enthusiasm is not optional — it is the medium
- slow down by 20%, especially at the start
- find a nodder, maintain eye contact with the room
- questions are success signals — welcome them

The one-sentence test

Before presenting: can you state your key idea in one clear sentence?

If yes — you are ready to build the talk. If no — spend more time on understanding before building slides.

> "Be yourself, but on purpose." — Every talk is a performance. Prepare the performance so the ideas come through.

Image Credits & References

Peyton Jones, Simon (2016). How to Write a Great Research Paper. https://www.microsoft.com/en-us/research/academic-program/write-great-research-paper/

Peyton Jones, Simon (2016). How to Give a Great Research Talk. https://www.microsoft.com/en-us/research/academic-program/give-great-research-talk/